LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #12 –

Healthcare & Telemedicine

Blood Bank Inventory & Emergency Donor Matcher

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SECTION: B

Github link:

https://github.com/anusha-g22/Blood-Bank-Emergency-Donor-Matcher

Actors:

1. **Emergency Requester**
2. **Blood Bank Manager**

Functional Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow an Emergency Requester to submit an emergency blood request specifying blood group, required units, emergency location, and urgency.	High	Pass if a valid emergency request is successfully recorded with all required details.	The system needs accurate emergency request information to process the blood requirement.	Ensure all mandatory request fields are validated.
FR-002	Functional	The system shall check	High	Pass if the system	The system must	Inventory quantity

		the current blood bank inventory for compatible blood units matching an emergency request.		displays the available compatible blood units and quantities for a valid request.	determine whether the required blood is already available.	should be updated after allocation.
FR-003	Functional	The system shall identify eligible donors with compatible blood groups within a 10 km radius when sufficient blood is unavailable.	High	Pass if only eligible donors with compatible blood groups within 10 km are selected.	Nearby compatible donors can help meet critical blood shortages quickly.	Donor eligibility and location must be verified.
FR-004	Functional	The system shall send emergency SMS notifications to compatible eligible donors identified by the donor matching process.	High	Pass if compatible donors receive the emergency notification within 30 seconds of the emergency flag.	Rapid notification is necessary during emergency blood shortages.	Notification delivery status should be recorded
FR-005	Functional	The system shall allow the Blood Bank Manager to add, update, allocate, and remove blood units while tracking	High	Pass if inventory changes are accurately recorded and expired blood units cannot be allocated.	Accurate inventory management prevents incorrect allocation and use of expired blood.	Expiry date must be checked before allocation.

		their expiry dates.				

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments:
NFR-001	Non functional	The inventory ledger shall maintain transactional consistency and prevent simultaneous allocation of the same blood unit.	High	Pass if concurrent allocation tests confirm that one blood unit cannot be allocated to multiple requests.	Double allocation can cause incorrect inventory records and serious operational errors.	Test using simultaneous allocation requests.
NFR-00	Non functional	The system shall process emergency donor matching and notification requests within 30 seconds under normal operating conditions.	High	Pass if at least 95% of simulated emergency requests result in donor notifications within 30 seconds.	Emergency situations require rapid identification and notification of donors.	Performance should be tested under simulated emergency load.

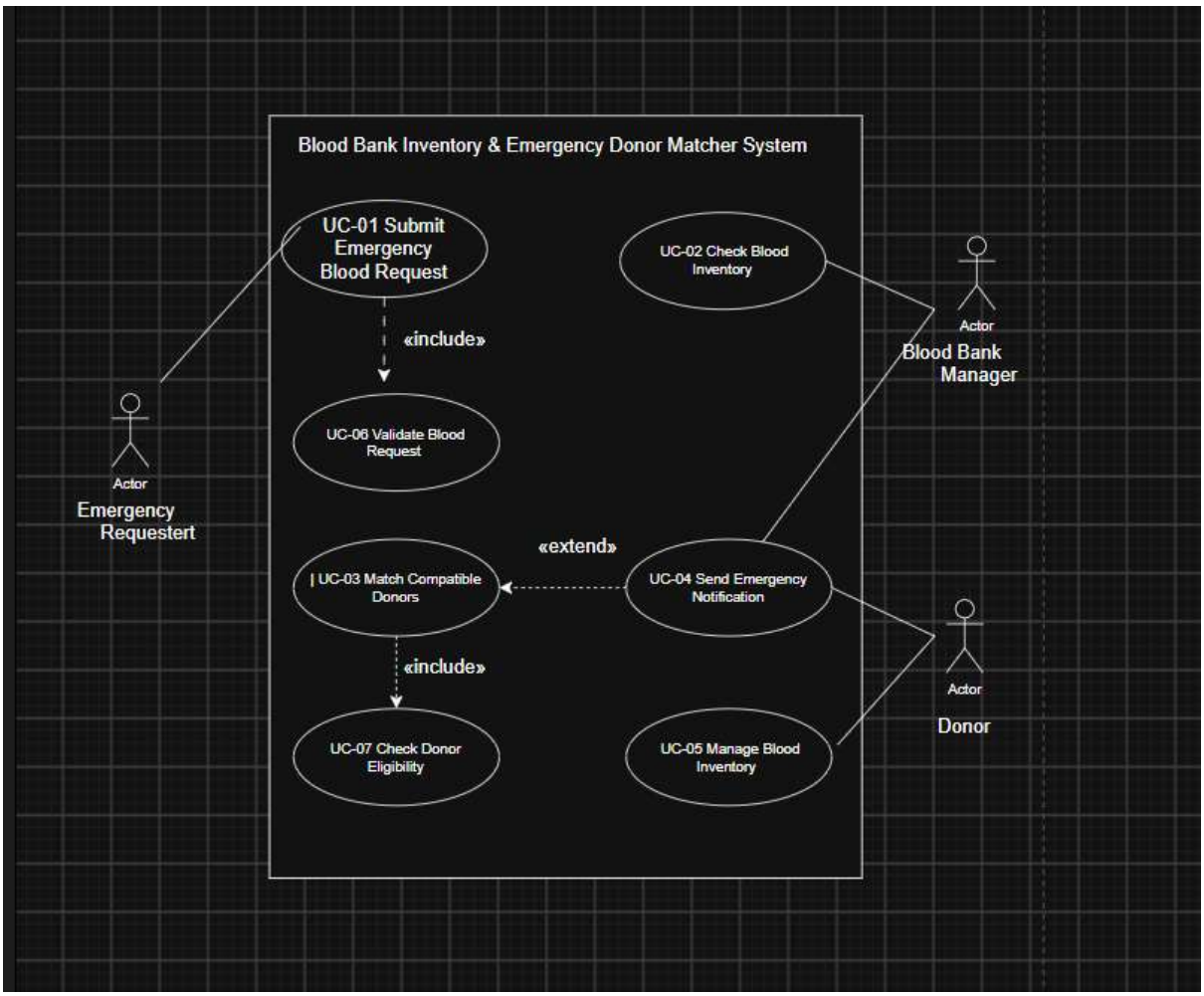
3. UMLUse-Case Diagram

Actor 1
Emergency Requester
Requests blood during an emergency.

Actor 2
Blood Bank Manager
Manages inventory and emergency requests.

Actor 3
Donor
Receives emergency notifications and can respond to the request.

This is reasonable because the system sends notifications to donors, so the **Donor directly interacts with the system**.



Use Case Flow: Emergency Blood Request

Use Case: UC-01 – Submit Emergency Blood Request

Primary Actor: Emergency Requester
Supporting Actors: Blood Bank Manager, Donor

Main Success Scenario
Emergency Requester selects the "Submit Emergency Blood Request" option from the system. The system displays a form to enter blood group, required units, emergency location, and urgency.
Emergency Requester enters a valid blood group, required number of units, and emergency location.

The system validates the entered request details.
The system checks the blood bank inventory for compatible blood units.
Sufficient compatible blood units are available in the inventory.
The Blood Bank Manager allocates the required blood units to the emergency request.

The system updates the blood inventory and records the allocation.
The system displays the successful allocation status to the Emergency Requester.
The emergency blood request is marked as fulfilled.
Use case ends successfully.

Alternate Flow
5a. Insufficient Blood Available

5a1. The system determines that sufficient compatible blood units are not available in the inventory.

5a2. The system searches for eligible donors with a compatible blood group within a 10 km radius of the emergency location.

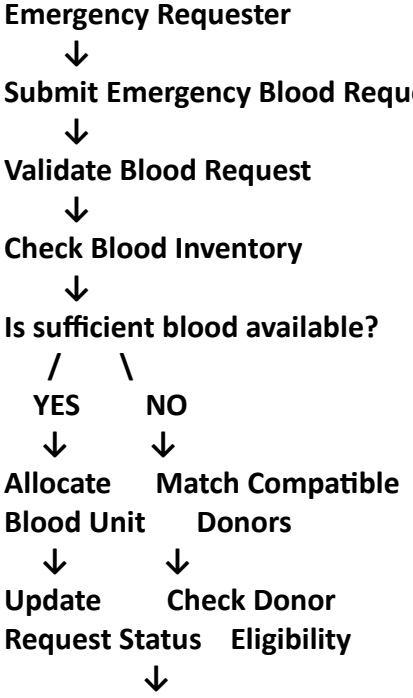
5a3. The system checks the eligibility of the identified donors.

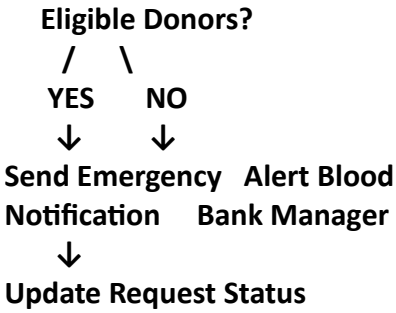
5a4. If eligible donors are found, the system sends emergency SMS notifications to the matching donors.

5a5. The system records the notification status and updates the emergency request as "Awaiting Donor Response."

5a6. If no eligible donors are found, the system alerts the Blood Bank Manager about the shortage.

5a7. The use case ends with the request marked as "Blood Shortage – Further Action Required."




Eligible Donors?
YES
NO
Send Emergency Notification
Alert Blood Bank Manager
Update Request Status